

Capstone Proposal

Methods and Data

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Data Processing and Analysis

Software and Packages

All data processing, analysis, and modeling will be conducted using free open source software and packages to maximize accessibility and reproducibility. The majority of data manipulation and processing will be performed in the R programming language using the tidyverse suite of packages (Wickham et al. 2019; R Core Team 2024). Any data operations that need to be “spatially aware” will be performed using the Simple Features package (Pebesma 2018).

Analysis will be scripted and ran in Positron, an open-source integrated development environment developed by Posit that supports R, Python, scripts, notebooks, integrated consoles, along with interactive plots and data viewers.

Study Area

The study area for this project is the Pensacola Bay System which includes portions of Pensacola Bay, Escambia Bay, East Bay and Blackwater Bay. Boundaries of this study area were achieved by accessing Florida Department of Protections (FDEP) Waterbod IDs feature layer hosted on ArcGIS Online (FDEP 2005). ArcGIS allows direct access to hosted feature services via several formats. To avoid any potential compatibility issues, the GeoJSON format was chosen as this format is non-proprietary. Once retrieved, the dataset was filtered to include only relevant WBIDs (WBID = 548) and then combined to create a singular study area polygon. Creating the study area polygon is an important first step as it will be needed to extract relevant environmental data from much larger datasets in the coming data gathering.

```
study_area <- get_study_area()

tm_shape(study_area) +
  tm_polygons(fill_alpha = 0, col = "red", lwd = 2) +
  tm_basemap("Esri.WorldStreetMap") +
  tm_layout(inner.margins = c(0.2, 0.2, 0.2, 0.2))
```

Water Quality Portal Data

Water quality data is widely available for the entire United States via the EPA's Water Quality Portal (WQP) which allows for a wide range of data querying for any data as it relates to Water Quality Monitoring programs (USGS (2026)). For this analysis, site data for Escambia and Santa Rosa County was queried using the WQP API then the study area polygon was used to extract only the relevant sites:

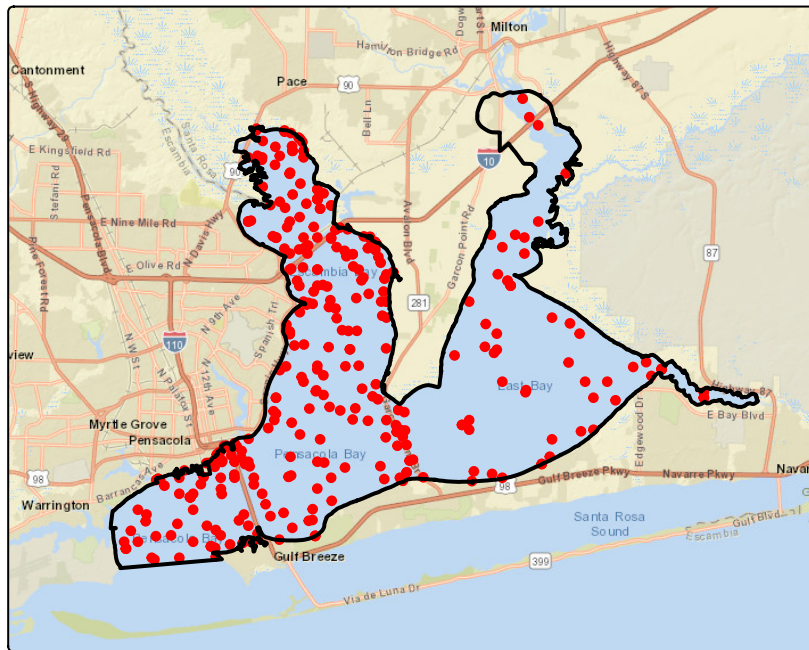


Figure 1: Pensacola Bay System with monitoring sites

Water Quality Data

Water quality data was selected based off of availability and relevance within literature. Each parameter was filtered to include just those measurements

```

salinity <- get_wq_data(
  param = "Salinity",
  from_date = "2005-01-01",
  to_date = "2025-01-01",
  sites = pbs_sites,
  output_name = "salinity",
  force = FALSE
)

water_temp <- get_wq_data(
  param = "Temperature, water",
  from_date = "2005-01-01",
  to_date = "2025-01-01",
  sites = pbs_sites,
  output_name = "water_temp",
  force = FALSE
)

do_mgL <- get_wq_data(
  param = "Dissolved oxygen (DO)",
  from_date = "2005-01-01",
  to_date = "2025-01-01",
  sites = pbs_sites,
  output_name = "do_mgL",
  force = FALSE
)

chl_a_mgL <- get_wq_data(
  param = "Chlorophyll a, corrected for pheophytin",
  from_date = "2005-01-01",
  to_date = "2025-01-01",
  sites = pbs_sites,
  output_name = "chl_a_mgL",
  force = FALSE
)

turbidity <- get_wq_data(
  param = "Turbidity",
  from_date = "2005-01-01",
  to_date = "2025-01-01",
  sites = pbs_sites,
  output_name = "turbidity",

```

```
force = FALSE
)
```

Table 1: Preliminary Environmental Variables Considered for the Eastern Oyster SDM

Variable	Obtained? (Y/N)	Data Type	Total Records	Data Source
Salinity	Y	Points	6994	National Water Quality Monitoring Council (2026)
Water temperature	Y	Points	7161	USGS (2026)
Substrate / bottom type	Y	Raster	NA	(ppbep?)
Water depth / bathymetry	N	Raster	National Centers for Environmental Information (2018)	NA
Dissolved oxygen	Y	Points	6261	USGS (2026)
Chlorophyll-a	Y	Points	1847	USGS (2026)
Oyster bed locations	NA	NA	FWC-FWRI (2026)	NA

```
# process wq data
```

```
sal <- proc_wq_data(salinity)
do <- proc_wq_data(do_mgL)
chla <- proc_wq_data(chl_a_mgL)
turb <- proc_wq_data(turbidity)
temp <- proc_wq_data(water_temp)
```

```
wq_summary <- bind_rows(
  sal %>% st_drop_geometry() %>% mutate(parameter = "Salinity (ppt)",
  do %>% st_drop_geometry() %>% mutate(parameter = "Dissolved Oxygen (mg/L)",
  chla %>% st_drop_geometry() %>% mutate(parameter = "Chlorophyll-a (µg/L)",
  turb %>% st_drop_geometry() %>% mutate(parameter = "Turbidity (FTU)",
  temp %>% st_drop_geometry() %>% mutate(parameter = "Temperature (°C)")
)
```

```
ggplot(wq_summary, aes(x = mean)) +
  geom_histogram(
```

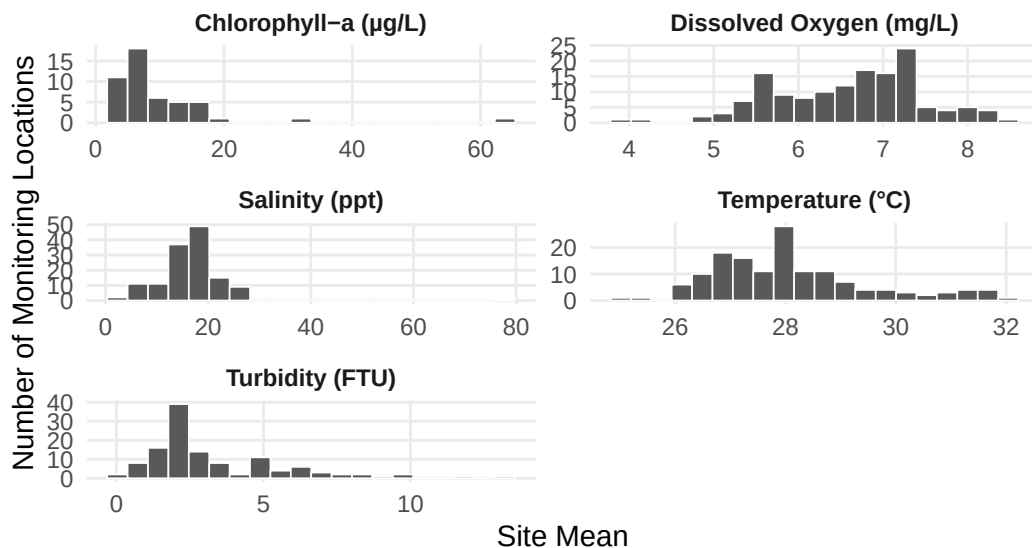
```

    bins = 20,
    color = "white",
    linewidth = 0.3
  ) +
  facet_wrap(
    ~ parameter,
    scales = "free",
    ncol = 2
  ) +
  labs(
    title = "Distribution of Warm-Season Water Quality Conditions",
    subtitle = "Site-level means for monitoring locations in the Pensacola Bay System",
    x = "Site Mean",
    y = "Number of Monitoring Locations"
  ) +
  theme_minimal(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 14),
    plot.subtitle = element_text(size = 10),
    strip.text = element_text(face = "bold"),
    panel.grid.minor = element_blank()
  )

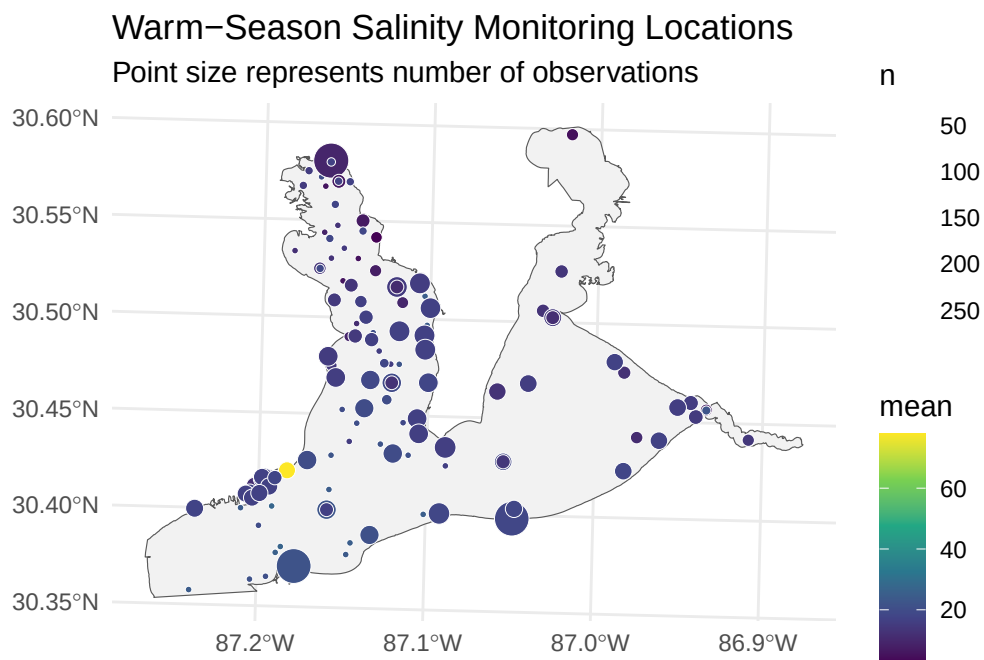
```

Distribution of Warm-Season Water Quality Conditions

Site-level means for monitoring locations in the Pensacola Bay System



```
# map points for each variable
ggplot() +
  geom_sf(data = study_area, fill = "grey95") +
  geom_sf(data = sal, aes(size = n, fill = mean),
    shape = 21,
    color = "white",
    stroke = 0.25
  ) +
  scale_fill_viridis_c(
    option = "viridis"
  ) +
  labs(
    title = "Warm-Season Salinity Monitoring Locations",
    subtitle = "Point size represents number of observations"
  ) +
  theme_minimal()
```

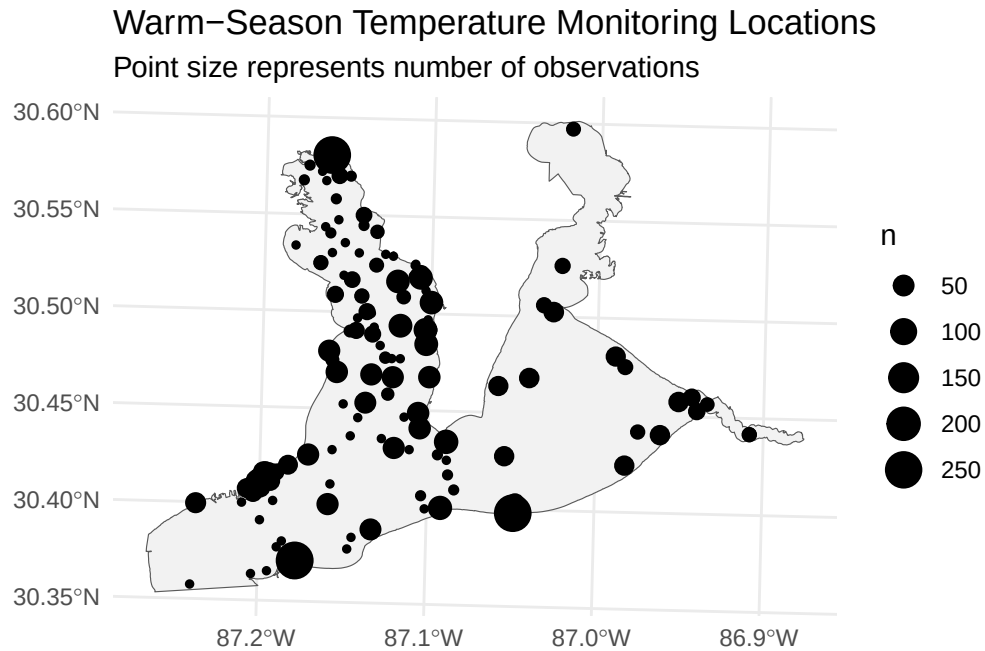


```
ggplot() +
  geom_sf(data = study_area, fill = "grey95") +
  geom_sf(data = temp, aes(size = n)) +
  labs(
    title = "Warm-Season Temperature Monitoring Locations",
```

```

    subtitle = "Point size represents number of observations"
  ) +
  theme_minimal()

```



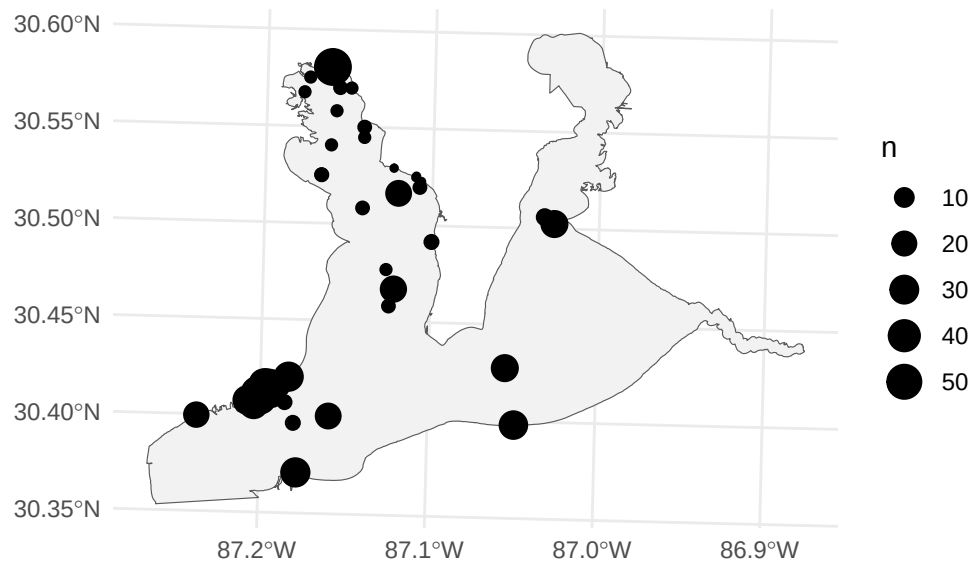
```

ggplot() +
  geom_sf(data = study_area, fill = "grey95") +
  geom_sf(data = chla, aes(size = n)) +
  labs(
    title = "Warm-Season Chlorophyll-a Monitoring Locations",
    subtitle = "Point size represents number of observations"
  ) +
  theme_minimal()

```

Warm-Season Chlorophyll-a Monitoring Locations

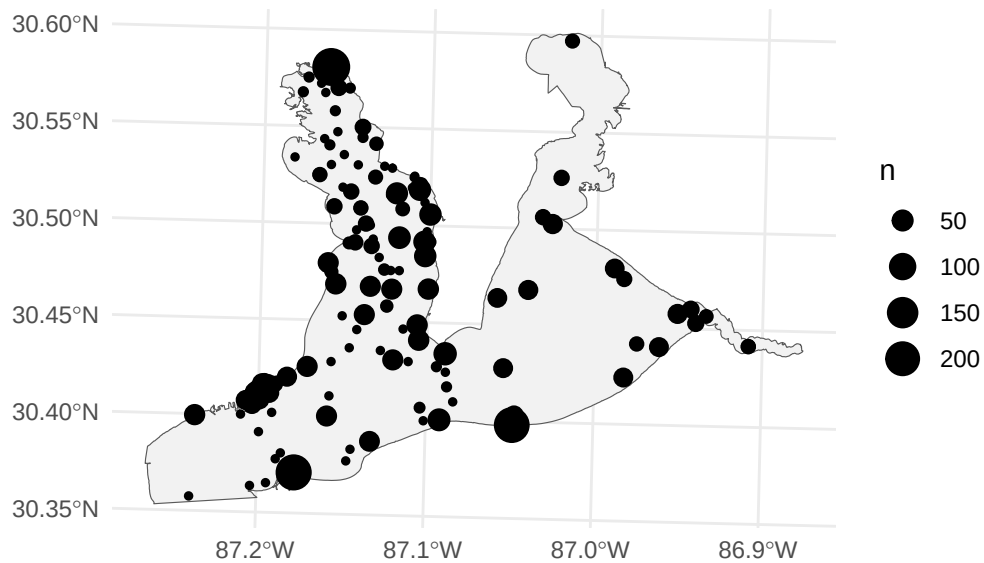
Point size represents number of observations



```
ggplot() +  
  geom_sf(data = study_area, fill = "grey95") +  
  geom_sf(data = do, aes(size = n)) +  
  labs(  
    title = "Warm-Season Dissolved Oxygen Monitoring Locations",  
    subtitle = "Point size represents number of observations"  
  ) +  
  theme_minimal()
```

Warm-Season Dissolved Oxygen Monitoring Locations

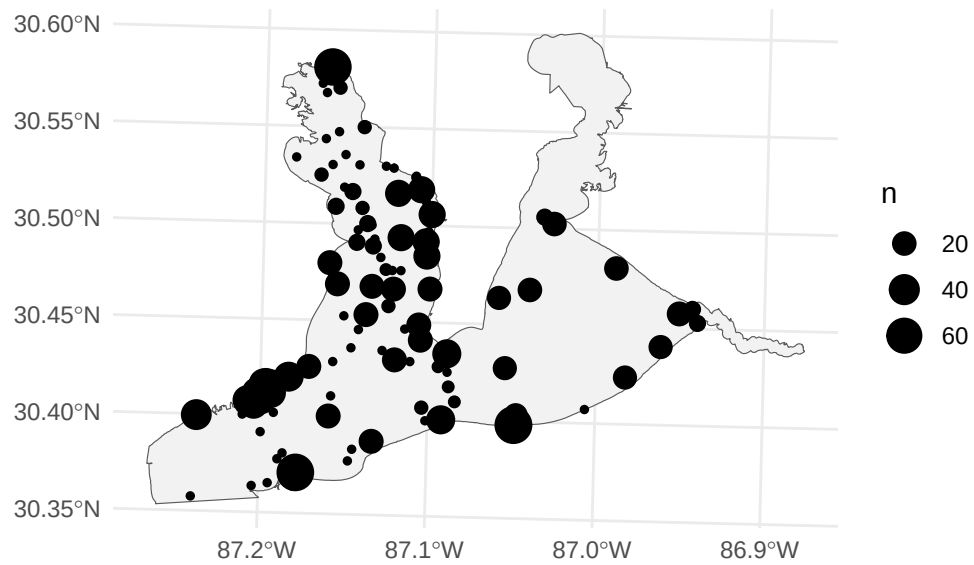
Point size represents number of observations



```
ggplot() +  
  geom_sf(data = study_area, fill = "grey95") +  
  geom_sf(data = turb, aes(size = n)) +  
  labs(  
    title = "Warm-Season Turbidity Monitoring Locations",  
    subtitle = "Point size represents number of observations"  
  ) +  
  theme_minimal()
```

Warm-Season Turbidity Monitoring Locations

Point size represents number of observations



```

wq_summary_table <- wq_summary %>%
  group_by(parameter) %>%
  summarise(
    stations = n(),
    observations = sum(n),
    overall_mean = mean(mean, na.rm = TRUE),
    overall_sd = sd(mean, na.rm = TRUE),
    min = min(mean, na.rm = TRUE),
    q1 = quantile(mean, 0.25, na.rm = TRUE),
    median = median(mean, na.rm = TRUE),
    q3 = quantile(mean, 0.75, na.rm = TRUE),
    max = max(mean, na.rm = TRUE),
    first_date = min(first_date, na.rm = TRUE),
    last_date = max(last_date, na.rm = TRUE),
    .groups = "drop"
  )

kable(
  wq_summary_table,
  caption = "Summary Statistics for Warm-Season Water Quality Data",
  col.names = c(
    "Parameter",
    "Stations",
    "Observations",
    "Mean",
    "SD",
    "Min",
    "Q1",
    "Median",
    "Q3",
    "Max",
  )
)

```

```
    "First Sample",
    "Last Sample"
),
align = c("l", "c", "c", "c"),
digits = 2,
booktabs = TRUE
)
```

Table 2: Summary Statistics for Warm-Season Water Quality Data

Parameter	Sta- tions	Observa- tions	Mean	SD	Min	Q1	Me- dian	Q3	Max	First Sample	Last Sample
Chlorophyll-a (µg/L)	48	754	9.74	9.70	3.39	5.14	6.21	11.79	63.80	2005-05-03	2024-10-17
Dissolved Oxygen (mg/L)	145	3072	6.59	0.87	3.90	5.82	6.75	7.21	8.48	2005-05-03	2024-10-17
Salinity (ppt)	135	3401	16.55	7.35	2.31	14.02	16.54	18.91	77.85	2005-05-03	2024-10-17
Temperature (°C)	145	3442	28.11	1.43	25.01	27.05	27.92	28.74	32.04	2005-05-03	2024-10-17
Turbidity (FTU)	122	1563	3.33	2.41	0.03	1.85	2.35	4.65	13.22	2005-05-10	2024-10-17

Spatial interpolation

Points are interpolated into prediction rasters using inverse path distance weighting (IPDW) (package `idpw`). Blah blah blah blah

References

- FDEP. 2005. “Waterbody IDs (WBIDs).” Geospatial Open Data; Florida Department of Environmental Protection. <https://geodata.dep.state.fl.us/datasets/FDEP::waterbody-ids-wbids/about>.
- FWC-FWRI. 2026. “Oyster Beds in Florida.” Geospatial Open Data; Florida Fish; Wildlife Conservation Commission, Fish; Wildlife Research Institute. <https://geodata.myfwc.com/datasets/myfwc::oyster-beds-in-florida/about>.
- National Centers for Environmental Information. 2018. “NOAA NOS Estuarine Bathymetry - Pensacola Bay (G130).” <https://doi.org/10.7289/V5XK8CWK>.
- National Water Quality Monitoring Council. 2026. *Water Quality Portal*. <https://doi.org/10.5066/P9QRKUVJ>.
- Pebesma, Edzer. 2018. “Simple Features for R: Standardized Support for Spatial Vector Data.” *The R Journal* 10 (1): 439–46. <https://doi.org/10.32614/RJ-2018-009>.
- R Core Team. 2024. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. <https://www.R-project.org/>.
- USGS, USDA, USEPA. 2026. “Water Quality Portal.” U.S. Geological Survey; U.S. Environmental Protection Agency; U.S. Department of Agriculture. <https://www.waterqualitydata.us/>.
- Wickham, Hadley, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D’Agostino McGowan, Romain François, Garrett Golemund, et al. 2019. “Welcome to the tidyverse.” *Journal of Open Source Software* 4 (43): 1686. <https://doi.org/10.21105/joss.01686>.